Appendix: Data Handling and Model Choices

MGT-403 Data Project · Where Should We Open Our Next Daycare?

Team: Winston Park, Ayush Rajpal, Tobin Gedstad, Wen Ding, Alfred Mordecai

A1. Sources

Source	What we used	Level	Years
DOL National Database of Childcare Prices (NDCP2022.xlsx)	MCPRESCHOOL: median full-time weekly price, center-based, preschool (36–54 months); MCINFANT and MCTODDLER for descriptives	County (5-digit FIPS)	2015–2022
BLS OEWS metropolitan & nonmetropolitan area files (oesmYYma.zip)	A_MEAN (mean annual wage) and H_MEAN for SOC 39-9011 Childcare Workers	OEWS area (7-character code)	May 2015–2022
Census ACS 5-year API (acs/acs5)	B19301, B01001, B23008, B25064, B15002, B23022 (codes in the assignment)	County	Vintages 2015–2022
Course crosswalk county_oews_crosswalk.csv	Year-specific county → OEWS area and area title	County × year	2015–2022
BLS CPI-U, CUUR0000SA0	Annual averages to convert to 2022 dollars	National	2015–2022
NDCP Technical Report (Sept 2024), Appendix B	Imputation codes by state and year (transcribed to data/ndcp_appendixB_imputations.csv)	State × year	2015–2022
Census cartographic boundary files (cb_2022, 20m)	County and state outlines for the map only	County	2022

Year	2015	2016	2017	2018	2019	2020	2021	2022
CPI-U	237.017	240.007	245.120	251.107	255.657	258.811	270.970	292.655
Multiplier to 2022 $	1.2347	1.2194	1.1939	1.1655	1.1447	1.1308	1.0800	1.0000

A2. Geography, identifiers and merges

- **Identifiers kept as text.** NDCP stores FIPS as an integer, so we restored the 5-character zero-padded code. OEWS area codes were zero-padded to 7 characters before joining, and the crosswalk was read with every column as text.
- **Wages to counties.** OEWS wages were joined to counties on [year, oews_area] using the year-specific crosswalk, and metro area titles came from the same file. For 2015–2017 the large metros are split into divisions in the MSA file; the whole-metro rows come from the aMSA file. No year × area appears twice.
- **Metro sample.** We kept counties whose OEWS area begins with "00" and dropped nonmetropolitan areas. Two retired FIPS codes in the crosswalk (12025, 51515: old Dade County FL and Bedford city VA) never appear in the ACS and would double-count their metro, so they were dropped.
- **Missing wages.** BLS suppresses a few area wages ("*"), which leaves 0.4–1.6% of metro county-years without a wage, in line with the roughly 1% the assignment says to expect. Areas suppressed in 2022: Arecibo, PR, Bremerton-Silverdale, WA, Columbus, IN, Fond du Lac, WI, Gadsden, AL, Guayama, PR, Homosassa Springs, FL, Janesville-Beloit, WI, Madera, CA, Mayaguez, PR, Pine Bluff, AR, Yuma, AZ.
- **Connecticut 2022.** The 2022 ACS reports Connecticut's nine planning regions instead of its eight counties, so the NDCP/crosswalk county codes have no ACS match that year and fall out of the 2022 analysis dataset. We did not reallocate planning-region data to counties.
- **Prices to counties.** NDCP was merged 1:1 on [FIPS, year]; every metro county-year matched. Puerto Rico municipios are metro counties in the crosswalk and have NDCP prices only in 2021, so they are in the dataset

for that year. They never pass the screens.

A3. Variable construction

- **Price.** MCPRESCHOOL, the median weekly full-time price for center-based preschool, as the assignment asks. The 75th-percentile price (_75CPRESCHOOL) is not used; it would be a natural extension for gauging the premium segment.
- **ACS-derived variables** follow the assignment exactly: kids_under6_allpar_lf = B23008_004E + 010E + 013E; female_college_share = (B15002_032E + 033E + 034E + 035E) / B15002_019E; female_ftyr_share = B23022_029E / B23022_026E; under6_share and under6_careneed_share divide by total population. Negative ACS annotation codes (e.g. -666666666) were set to missing. We confirmed that each variable code has the same label in every vintage.
- **Real dollars.** Prices, wages, rents and income were multiplied by CPI_{2022}/CPI_t (table A1). Wages are scaled to \$1,000s and rents to \$100s so coefficients read as "\$/week per \$1,000 of wage" and "per \$100 of rent".
- **Analysis dataset.** A year × county row is kept only if the price and all ten predictors are non-missing (8,146 rows). The file is saved as output/analysis_dataset.csv.

A4. NDCP price quality: what the data really measure

NDCP county prices come from each state's market-rate survey (MRS). Two features matter for a county ranking.

- **Rate-zone prices.** Many states publish one price per rate zone, so different counties report an identical price. 65% of analysis county-years share their exact price with at least one other county in the state; in 2022 the median number of counties sharing a price is 4, and only 42% of counties have a price shared by two or fewer. Example: 10 Wisconsin metro counties (Brown, Dane, Eau Claire, Kenosha, La Crosse, Milwaukee, Racine, Rock, Waukesha, Winnebago) all report \$258.75 in 2022. In a zone, the lower-cost counties show a large positive residual purely because the price is averaged across the zone. That is a measurement artifact, not real pricing power. Our county-specific-price screen (price shared by at most 2 counties) removes this; such counties go to the verify-first list instead.
- **Imputations (Technical Report, Appendix B).** Code 5 = estimate for a year between surveys (linear interpolation, or extrapolation up to two years after the last survey); 4 = one of the 50th/75th percentiles estimated from the other with a lognormal model (NDCP does not say which, so the median may be modeled); 3 = county prices imputed from state-level data; 6 = propensity-score-match estimates (California, North Carolina and Virginia in 2022); 2 = conversion to a weekly price. We exclude codes 3 and 6 in 2022 from the recommendation. We keep codes 4 and 5, which together apply to 64% of 2022 analysis counties, so dropping them would remove most of the country; instead we test sensitivity to code 5 ("survey years only") and show the evidence behind each pick in the analysis document. 7 of the 10 picks' 2022 prices are code-5 estimates. Several states that Appendix B lists as excluded ("NA") in a year nonetheless have prices in the 2022 workbook (e.g. AK, KS, KY, LA, MI, MT, SC, WV in 2022). We treat those as undocumented, not imputed.
- **The *_flag variables** (e.g. MCPreschool_flag = 1/2/3) record how detailed age brackets were combined into the preschool group (all values equal / mode / highest mode). They are not imputation markers, so we do not use them as quality screens.

A5. Model choices

- **Levels, not logs, for the main model.** A dollar premium per week reads directly as extra revenue per child relative to the wage-and-rent benchmark (it is not a profit margin), and levels match the pre-analysis regressions. The log model fits almost identically out of sample (R^2 0.602 vs 0.598) and is a robustness check.

- **Year-by-year benchmark models** rather than one pooled model, so the benchmark adjusts if the price-cost relationship shifts (e.g. 2020–2021). All years use real 2022 dollars, so premiums are comparable across years.
- **No demand variables in the benchmark model.** The assignment defines attractiveness as a high price given wages and rents. Putting income or education in the benchmark would absorb exactly the willingness to pay we want to find. We study demand characteristics separately, in part (c).
- **No state fixed effects in the main model.** Between-state price differences can be real pricing power (or regulation that raises cost). 43% of premium variance is between states, so we report a within-state version as robustness (2022 R^2 rises to 0.80). All 10 picks have a positive within-state premium, and 9 rank first or second in their state.
- **Inference.** Wages vary only at the metro level and nearby counties share shocks, so standard errors are clustered by OEWS area. Cross-validation holds out whole metro areas for the same reason.
- **Screens and thresholds.** At least 3 of 4 recent years, including 2022, so the score is current; positive in every observed year, for persistence; zone size ≤ 2, for a county-specific price; ≥ 2,000 under-6 children with working parents, so a ~150-seat center would need under 8% of local demand; one county per metro, to diversify across labor and rental markets.
- **Back-test design.** Same rules, applied with 2015–2018 scores and 2018 eligibility (market size, zone, codes), and evaluated on the 2022 premium, a 4-year gap that exceeds NDCP's maximum 2-year extrapolation.

A6. Robustness of the top 10

Alternative rule	Overlap with baseline	Dropped	Added
Log model (% premium)	6/10	King, Washington; Montgomery, Maryland; Worcester, Massachusetts; Washtenaw, Michigan	Winnebago, Illinois; Cascade, Montana; Missoula, Montana; Raleigh, West Virginia
2022 premium only	8/10	Worcester, Massachusetts; Washtenaw, Michigan	Anchorage, Alaska; Wood, West Virginia
Survey years only	9/10	Washtenaw, Michigan	Anchorage, Alaska
Within-state premium (state FE)	9/10	Worcester, Massachusetts	Multnomah, Oregon

6 counties (Stearns, Norfolk, McLean, Howard, DuPage and Champaign) stay in the top 10 under every rule. Worcester and Washtenaw are the least robust; the next in line (Missoula, Montana; Multnomah, Oregon; Anchorage, Alaska) are natural substitutes.

Back-test picks (chosen with 2015–2018 data)

County	Metro	2015–18 score	2022 premium
Arlington, Virginia	Washington-Arlington-Alexandria, DC-VA-MD-WV	+$109	+$201
Multnomah, Oregon	Portland-Vancouver-Hillsboro, OR-WA	+$107	+$38
Norfolk, Massachusetts	Boston-Cambridge-Nashua, MA-NH	+$103	+$116
Benton, Oregon	Corvallis, OR	+$96	+$36
Washington, Pennsylvania	Pittsburgh, PA	+$75	no 2022 price
Fairfield, Connecticut	Bridgeport-Stamford-Norwalk, CT	+$74	no 2022 price
Story, Iowa	Ames, IA	+$73	+$8
Chester, Pennsylvania	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	+$68	no 2022 price
Mercer, Pennsylvania	Youngstown-Warren-Boardman, OH-PA	+$64	no 2022 price
Lane, Oregon	Eugene, OR	+$63	+$23

A7. Limitations

- NDCP measures posted market-rate-survey prices, not transaction prices or occupancy; a high price with low enrollment would not be profitable.
- OEWS wages are metro averages for all childcare workers, not preschool teachers at centers, and exclude benefits.
- Median gross rent is residential, not commercial.
- ACS 5-year estimates average five years, so 2022 values describe 2018–2022.
- The benchmark model is descriptive. Coefficients are associations used to build a benchmark price, not causal cost pass-through.

A8. Reproducibility and AI assistance

All code and data are public at **https://github.com/ayush-rajpal/mgt403-daycare-location**. **python run_all.py** runs every step in order and then an independent check that recomputes the key results from the raw files:

Step	Script (src/)	Produces
1	step1_ps4_prices.py	PS4 tables and figures (outputs/ps4)
2	step2_ps4_report.py	PS4 report PDF
3	step3_ps5_cost_panel.py	Year × county metro panel: OEWS wages + ACS + crosswalk (outputs/ps5)
4	step4_ps5_report.py	PS5 report PDF
5	step5_final_analysis.py	Analysis dataset, regressions, scores, figures, outputs/final/results.json
6	step6_final_documents.py	This appendix and the pre-analysis / final-analysis PDF
7	step7_final_deck.js	Daycare_Expansion_Deck.pptx (6 slides)
check	checks/verify_results.py	Independent recomputation of every key number

Python 3.14 with pandas, numpy, statsmodels, scipy, matplotlib, pyshp and reportlab; Node.js with pptxgenjs. The work was done with an AI coding assistant (Claude Code) following the team's prompt: "complete the assignment… we have to submit the pre-analysis work, a short slide deck summarizing the findings, and any appendices describing data cleaning choices or model choices; for the slides use Yale blue and blue shades with a white background." The assistant reused the team's PS4/PS5 data and code, wrote the final-project steps 5–8, and checked every figure and slide by rendering it. Every statistic in the documents and slides is computed by the code (from outputs/final/results.json or the output CSVs), not typed by hand, and all of them were re-verified with an independent recomputation from the raw files.